ETHAN DO

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EDUCATION

Princeton University **September 2022 - May 2026**
Bachelor's, Computer Science

- Relevant Coursework: Data Structures and Algorithms, Full Stack Development, Data Science, Functional Programming, Database Management, Computer Systems and Networks, Startup Research, Machine Learning, Video Game Development, Algorithm Theory, Technology Policy and Law, Augmented Reality in AI Research

University of Hong Kong **January 2025 - May 2025**
Bachelor's, Computer Science

- Study abroad exchange semester through Princeton University

PROFESSIONAL EXPERIENCE

Fourth Age **New York City, New York**
Forward Deployed Engineer *June 2026 - Present*

- Designed a take-home technical assessment (a coffee-shop review app) with a 1-week build window, used monthly to evaluate 5+ candidates' Palantir Foundry skills across ontology design and data pipelines
- Performed ontology design and data cleaning/transforms linking 16 datasets, and built custom dashboards on the Palantir Foundry platform
- Supported asset management workflows for energy-sector clients, building Foundry-based data and reporting tooling

WIT Sports **New York City, New York**
Software Engineer Intern *May 2025 - August 2025*

- Built a fraud detection pipeline to uncover unauthorized posts of prize activations (e.g., car giveaways), preventing inflated contest entries and bot abuse
- Used Google Search API and Playwright/BeautifulSoup to scan for unapproved domains sharing activation links, filtering out known safe sites
- Deployed a headless crawler with Slack alerts and GitHub Actions scheduling for 30-minute scans, expanding fraud monitoring from 1 known site to 50+.

NashTech Global **Ho Chi Minh City, Vietnam**
Software Engineer Intern *June 2024 - August 2024*

- Built a .NET framework in C# to automate parsing and migration of XML-based applicant data from legacy systems into Azure SQL for King's College London
- Eliminated 15+ hours/week of manual data entry and reduced input errors by 80%, streamlining the school's transition to a modern tech stack
- Developed a lightweight internal UI with Bootstrap and Entity Framework 6 to ensure secure, reliable data migration.

PROJECTS

World Cup 2026 Bayesian Prediction Pipeline **July 2026 - August 2026**

- Built a hierarchical Bayesian Poisson goal-scoring model (NumPyro/NUTS) over 964 historical World Cup matches to simulate the live 2026 knockout bracket with posterior uncertainty across 40,000 tournament simulations
- Engineered a reproducible data pipeline (SQL-ready Parquet schemas) integrating historical results, live Elo ratings, and Kalshi prediction-market prices; validated the model against an Elo baseline on held-out 2018/2022 tournaments
- Built a market-pricing and trading strategy layer that blends model probabilities with de-vigged market prices (Kelly sizing, fee-adjusted edge thresholds) with a React dashboard for live scanning and position tracking

recipefy **January 2026 - June 2026**

- Developed a real-time AR cooking assistant using YOLO-based ingredient detection and LLM-generated recipe recommendations.
- Engineered a Flask + OpenCV pipeline with asynchronous API calls, motion detection, and stability locking for smooth live performance.
- Designed the system to support cuisine suggestions, step-by-step guided instructions, and gamified cooking experiences.

SKILLS

Skills: Java, Python, C#, SQL, C/C++, Haskell, Golang, JavaScript, HTML/CSS, Arm, Assembly, R
Frameworks and Libraries: React.js, Flask, Bootstrap, jQuery, AJAX, .NET, Entity Framework 6, NumPy
Tools & Platforms: Git, Agile, Google Cloud Platform, Microsoft Azure, Render, Cloudinary, Unity, Palantir Foundry
Databases: PostgreSQL, SQLite3
Creative Software: Adobe Photoshop, Adobe Premiere Pro, Adobe Lightroom